



DEPARTMENT OF
COMPUTER SCIENCE

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BSc in Computer Science and Engineering

AI-POWERED REQUIREMENTS ENGINEERING

A NEW APPROACH

MASTER IN COMPUTER SCIENCE AND ENGINEERING

NOVA University Lisbon

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ABSTRACT

In software development, requirements engineering plays a critical role in ensuring that software products meet stakeholder expectations and standards. However, the manual writing and thinking of requirements gathering, refinement, and validation introduces the risk of misunderstanding, incomplete requirements, and communication breakdowns between stakeholders and development teams.

Although AI techniques have been explored in various stages of software development, their application in automating and improving requirements engineering is still a new field of study. This thesis aims to explore the use of Generative AI (GAI) to automate the identification, clarification, and specification of requirements, ultimately improving the efficiency and accuracy of the RE process.

Concerning its contents, the abstracts should not exceed one page and may answer the following questions (it is essential to adapt to the usual practices of your scientific area):

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution/contribution?
4. What results (implications/consequences) from the solution?

Keywords: One keyword, Another keyword, Yet another keyword, One keyword more, The last keyword

RESUMO

Palavras-chave: Primeira palavra-chave, Outra palavra-chave, Mais uma palavra-chave,
A última palavra-chave

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INTRODUCTION

1.1 Motivation and Context

This first Chapter introduces the template and how it is organized. In Chapter 2 you can find some specific instructions on how to use this template. Chapter 3 shows some examples and give some hints on how to write your text. Please read these next Chapters carefully.

1.1.1 Motivation and Context

Did you learn how to drive by sitting by the wheel and throwing your car into the road? Most probably you did take your time *learning the rules* and *practicing* first! Likewise, it is not wise to throw yourself at the task of writing a thesis/dissertation in L^AT_EX without seriously considering the following recommendation!

Teste caso queira usar

1.1.2 Problem Definition

Problem definition

1.1.3 Objectives

Thesis objectives

1.1.4 Proposed Solution

Proposed Solution

NOVATHESIS TEMPLATE *USER'S MANUAL*

2.1 Introduction

This Chapter describes how to use the template. It is assumed that you have a working of $\text{\LaTeX}$, either local (in your own computer) or remote, and that you were able to generate a PDF for the default c

2.2 Quick Start

2.2.1 With a Local $\text{\LaTeX}$ Installation

Follow these steps to get started with a local $\text{\LaTeX}$ installation:

A SHORT L^AT_EX TUTORIAL WITH EXAMPLES

This Chapter aims at exemplifying how to do common stuff with L^AT_EX. We also show some stuff which is not that common! ;)

Please, use these examples as a starting point, but you should always consider using the *Big Oracle* (aka, [Google](#), your best friend) to search for additional information or alternative ways for achieving similar results.

3.1 Document Structure

ADDING SUPPORT TO A NEW SCHOOL (WORK IN PROGRESS)

My advice to customize the **novathesis** template to another School/University/Department/Degree is to browse the existing supported degrees to find one that is *close enough*, and depart from there!

The multitude of layouts supported by the **novathesis** template is based in a three-tier naming scheme, separated by slashes: University / School / Department-or-Degree. This three-tier naming scheme is also reflected in a three-tier directory (folder) structure in: `<project_root>a/NOVAthesiFiles/Schools/...` For example:

